

Unit 7 Pre-Test

Scatterplots & Two-Way Tables

Standards assessed: NC.8.SP.1–SP.4

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.SP.1 Describe patterns of association in bivariate data.
• *I can describe associations in scatterplots and name their key features.*

Match each vocabulary term to its definition, then answer each question.

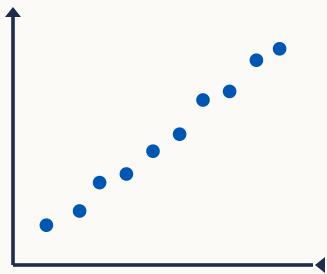
C ▼ scatterplot ✓	A. As one variable increases, the other tends to increase.
A ▼ positive association ✓	B. A data point that lies far away from the rest of the data.
D ▼ negative association ✓	C. A graph that plots paired data as points.
B ▼ outlier ✓	D. As one variable increases, the other tends to decrease.
E ▼ line of best fit ✓	E. A straight line drawn to model the trend in a scatterplot.
F ▼ cluster ✓	F. A group of data points bunched closely together.

1. Association

As study time increases, test scores increase. Positive or negative association?

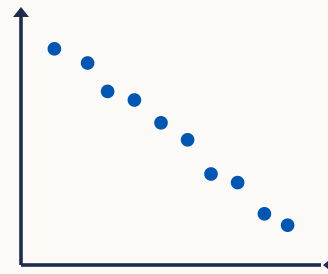
2. Association

As a car’s age increases, its value decreases. Positive or negative?



Answer:

answer: positive

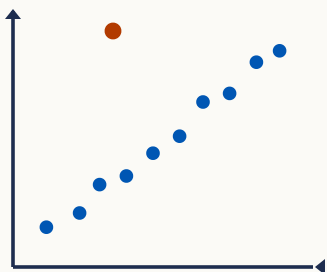


Answer:

answer: negative

3. Vocabulary

A point lying far from the pattern is called a(n) ____.

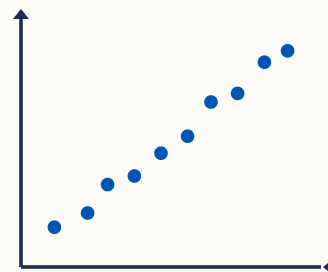


Answer:

answer: outlier

4. Vocabulary

A graph of paired data points is called a ____.

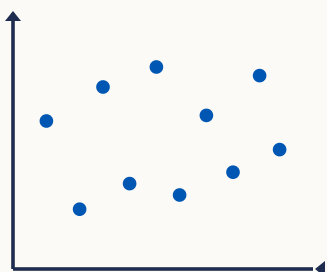


Answer:

answer: scatterplot

5. Association

If x and y show no pattern at all, there is ____ association.

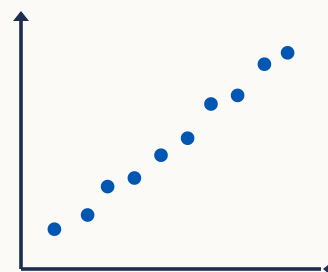


Answer:

answer: no

6. Vocabulary

A line drawn to model the trend of a scatterplot is the line of ____.



sketch the line of best fit

Answer:

answer: best fit

SCORE 4.0 Target Content

NC.8.SP.3, NC.8.SP.4 Use a line of best fit to predict, and read two-way tables.

• *I can use a line of best fit to make predictions and read values from a two-way table.*

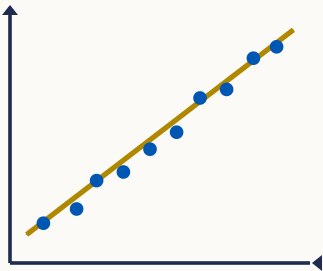
Use the line of best fit and the two-way table to answer each question.

Two-way table — use for the questions below

	Likes Pizza	Does Not	Total
Boys	12	8	20
Girls	15	5	20
Total	27	13	40

7. Predict

Line of best fit $y = 2x + 5$. Predict y when $x = 10$.

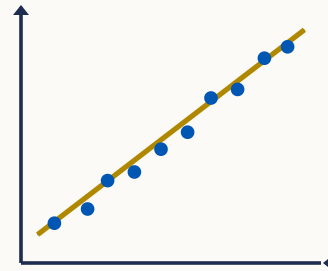


$y =$

answer: 25

8. Predict

Line of best fit $y = 2x + 5$. Predict y when $x = 0$.



$y =$

answer: 5

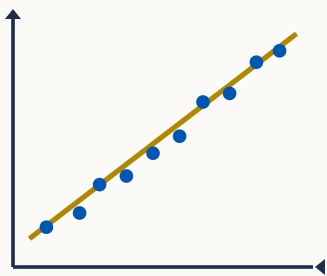
9. Interpret

In $y = 2x + 5$, by how much does y increase for each increase of 1 in x ?

10. Two-way table

From the table, how many students like pizza in all?

$=$



answer: 27

=

answer: 2

11. Two-way table

From the table, how many girls were surveyed?

=

answer: 20

12. Two-way table

From the table, how many students were surveyed in total?

=

answer: 40

SCORE **5.0** Advanced Content

NC.8.SP.4 Interpret relative frequencies and the meaning of slope in context.

• I can find relative frequencies from a two-way table and interpret slope in a real situation.

Use the two-way table above and each best-fit line to answer.

13. Relative freq.

From the table, what percent of boys like pizza?

=

answer: 60%

14. Relative freq.

From the table, what percent of girls like pizza?

=

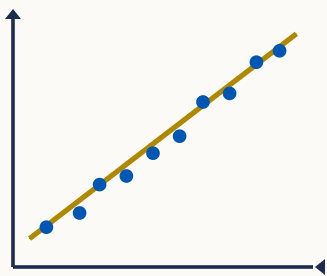
answer: 75%

15. Predict

Best-fit line for plant height: $y = 3x + 2$. Predict the height at $x = 6$ weeks.

16. Vocabulary

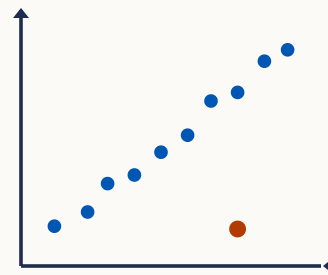
A scatterplot rises left to right, but one point sits far below the rest. That point



y =

answer: 20

is a(n) ____.

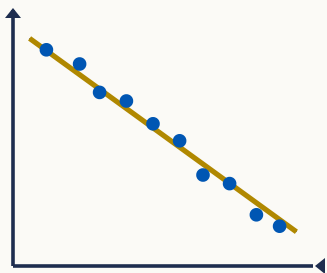


Answer:

answer: outlier

17. Interpret

Value vs. age: $y = -4x + 50$. By how much does the value drop each year?

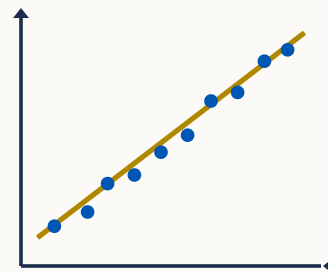


=

answer: 4

18. Predict

Line of best fit $y = 1.5x + 10$. Predict y when $x = 8$.



y =

answer: 22